WHEN GOVERNANCE CHANGES WHAT GETS COUNTED: MEASUREMENT VALIDITY FOR LLM AGENT EVALUATION

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ABSTRACT

Runtime validators have two roles in coupled agent models. They keep simulated agents within the behavioral rules that make the environment interpretable. We do not question that model-fidelity role. We study a different risk: an agent writes the condition that selects which of its own decisions enter a conditional score. We study that score: among decisions whose response reports high water scarcity, how often does the executed action increase demand?

The primary analysis compares validator-on and validator-off simulations across five seed-paired irrigation configurations. In the anchor cell, high-scarcity reports appear in 82.80% of no-validator decisions and 37.08% of validator-on decisions, a $+45.73$ pp contrast. The executed conditional increase contrast is $+25.90$ pp; the executed all-decision contrast is $+29.90$ pp. Across all five cells, executed all-decision contrasts range from $+20.26$ pp to $+36.57$ pp. The sample conditional-minus-all gap ranges from -16.06 pp to $+3.38$ pp; paired intervals exclude zero for three negative gaps and one positive gap, while one includes zero. Reweighting both arms to the same recorded world-state distribution leaves -2.53 pp at full coverage; the $+0.21$ pp finest grouping retains 43.4%/36.4% of decisions.

We also retain a secondary Flood check as a bounded cross-testbed comparison. Its different action ontology, feedback structure, and seed matching prevent it from serving as a second mechanism test.

These results show a measurement-validity risk: a governed-agent score can change because its measured population changes. Report the condition share, an independent world-state record, the action reading, and an all-decision rate beside any conditional score. Missing first responses prevent us from determining whether the agent would report the same condition in the same world; we make no deception claim.

1 INTRODUCTION

Evaluating an AI agent requires more than counting its actions. It also requires specifying which decisions enter the score. We call a score computed only over decisions that satisfy a specified condition a *conditional score*. Agent evaluations increasingly involve tool use, interaction, and feedback from an environment (Liu et al., 2024; Ma et al., 2024; Zhou et al., 2024; Kapoor et al., 2024; Xie et al., 2024; Xu et al., 2025). Runtime validators, components that check a proposed action before execution, can reject a response or request another one (Rebedea et al., 2023; Wang et al., 2026). In coupled human–environment models, keeping simulated behavior within its behavioral rules is also what makes later environment states interpretable (Khan et al., 2017; Lin et al., 2022; Streefkerk et al., 2023). The measurement problem arises when an agent-written field affects runtime handling and also selects the records used for evaluation.

Consider a water-management agent in a coupled reservoir simulation. At each yearly decision, it proposes an action and reports water scarcity and coping capacity, its reported ability to adapt. A validator checks the proposal against these reports and the simulator state. Under the studied rule, an increase can be rejected when both reported conditions are high or very high; high scarcity alone is only a warning. The validator can request a new response, or a default action executes after retry

exhaustion. If the evaluation also counts only decisions with a high-scarcity report, that response field participates in both enforcement and scoring. We call such a field an *agent-written condition*.

The arithmetic is simple. Suppose 80 of 100 decisions satisfy the agent-written condition, and 40 of those decisions take the action of interest. The conditional rate is 50%. After governance is enabled, suppose 40 decisions satisfy the condition and 10 of those decisions take the action. The rate is 25%. Both the numerator and denominator changed, so the two rates do not isolate a change in action behavior. In a coupled environment, the difference can propagate. An action changes a later state, that state becomes part of the next input, and later records may come from a different population. Figure 1 summarizes this dependency: the response supplies both the action and the condition, validation determines what executes, and the condition then selects records from the log. This is related to the statistical warning against conditioning on a variable changed by an intervention (Rosenbaum, 1984). Rosenbaum's warning usually concerns a covariate chosen by the analyst; here the evaluated system writes the conditioning field, so removing it changes the score rather than repairing it.

Agent evaluation now spans interaction, progress across multiple turns, and end-to-end outcomes rather than only single-turn accuracy (Kapoor et al., 2024; Liu et al., 2024; Ma et al., 2024; Xie et al., 2024; Xu et al., 2025). WebArena and SWE-bench check outcomes in websites and repositories, while OSWorld and TheAgentCompany extend evaluation to computer environments and consequential tasks (Zhou et al., 2024; Jimenez et al., 2024; Xie et al., 2024; Xu et al., 2025). ReAct, Reflexion, AutoGen, SWE-agent, ToolSandbox, and τ-bench make action–feedback loops, tool coordination, evolving state, and policy adherence part of evaluation (Yao et al., 2023; Shinn et al., 2023; Wu et al., 2023; Yang et al., 2024; Lu et al., 2025; Yao et al., 2025). AgentDojo and AgentIF add tool-mediated threats and explicit constraints, while NeMo Guardrails and AgentSpec show how runtime systems inspect or constrain responses (Debenedetti et al., 2024; Qi et al., 2025; Rebedea et al., 2023; Wang et al., 2026). Construct-validity and benchmark-rigor work asks whether a benchmark measures the behavior it claims to measure (Bean et al., 2025; Zhu et al., 2025). However, a specific case remains unresolved: the evaluated system can generate the condition that determines which of its own decisions enter the score. The system therefore helps generate its measured population before the evaluator receives the log. This turns a familiar conditioning warning into an agent-benchmark design risk with an auditable reporting protocol. Without that audit, a conditional rate can change through record selection while being read as a change in action behavior.

We use the coupled irrigation simulation as the primary empirical example of this problem. Paired validator-on and validator-off runs are matched by seed. We report the share of decisions carrying the agent-written condition, the conditional action rate using the executed action, the executed action rate over every decision, and descriptive reweighting by simulator-recorded world state. Flood is used only as a secondary descriptive boundary check; it is not a matched control, a mechanism-transfer test, or a causal comparison. We compare validator-on with validator-off because that is the contrast governed-agent evaluations report; the paper studies what this contrast contains, not a causal governance effect.

In the primary irrigation analysis, the anchor configuration used for the detailed composition analysis has a $+45.73\,\mathrm{pp}$ contrast in condition share (Table 1). Across all five irrigation configurations, the executed all-decision contrast remains positive, while the conditional-minus-all-decision gap ranges from $-16.06\,\mathrm{pp}$ to $+3.38\,\mathrm{pp}$. The conditional contrast is smaller in four sample cells and larger in one, so the conditional rate alone does not reveal how it differs from the all-decision result. Replacing the agent report with a simulator-recorded shortage condition is not an automatic repair: its anchor contrast is $+64.29\,\mathrm{pp}$. The next section defines the paired evaluation and rule inputs; the paper then reports the moving denominator, diagnoses world-state composition, compares all-decision rates, and closes with a reporting protocol and limitations.

Our contributions are threefold:

1. We identify a measurement-validity risk for runtime governance evaluations in which a condition authored in the response also selects the records used for scoring.

2. We diagnose the risk with paired simulations, independent world-state records, and a cross-testbed check. Descriptive alignment of recorded states greatly reduces the raw contrast. The executed all-decision contrast remains positive across all five irrigation cells, but the sample conditional-minus-all gap is negative in four cells and positive in one; paired intervals show that one negative direction remains uncertain.

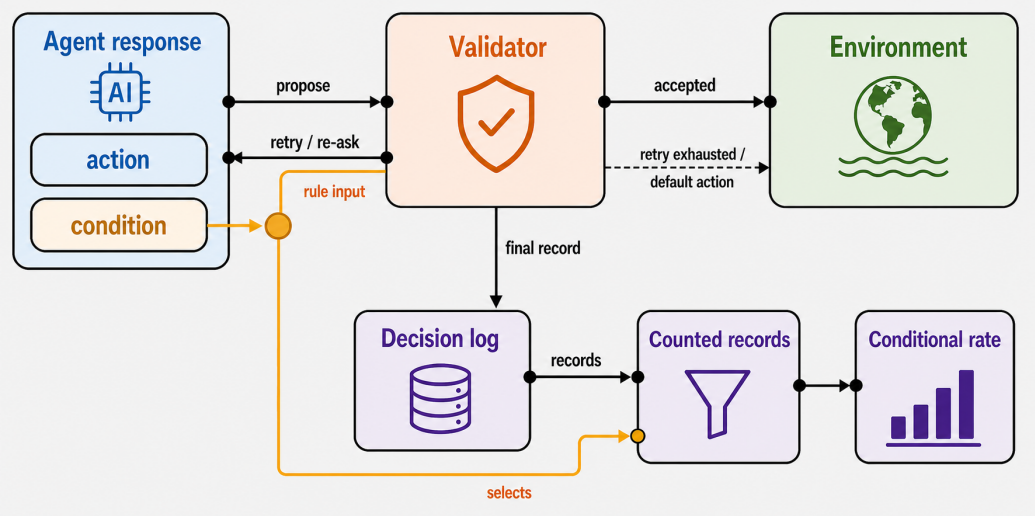


Figure 1: **How governance changes what gets counted.** The agent response contains an action and an agent-written condition. The validator reads both before execution: an accepted action reaches the environment, while a rejected proposal returns to the Agent response for a retry or, after retry exhaustion, a default action executes. The final executed action enters the decision log. The same condition selects the logged records used for the conditional rate. Colors mark roles: blue is agent output, orange is validation, green is the environment, violet is measurement, and amber is the shared condition; black arrows show flow. The displayed condition abstracts the primary scarcity report. The full rule also reads coping capacity, so high scarcity alone is not a block. The diagram is domain-neutral and does not compare irrigation with Flood. Irrigation-specific evidence appears in Figures 2–4, and Flood appears only in Table 3. The first response before a retry is omitted because it is absent from the frozen corpus. This is a conceptual diagram, not a data figure.

3. We give a minimal reporting protocol: report the condition share, action reading, all-decision rate, independent world state, and pre-retry response/message.

2 Evaluation Setting

2.1 Governance loop and rule inputs

In each simulated decision, the agent returns an action and two reports about its situation. A validator checks the proposal against those reports and the simulator's recorded state. For an error-level violation, it returns a validation message and asks for a fresh response, allowing up to three retries (four responses including the initial response). If all retries fail, a domain default executes. The released audit record stores one row per agent-year and uses the final executed action, denoted EXECUTED, for the primary statistics. The historical final-proposal reading, denoted LAST-PROPOSED, is retained only as a sensitivity analysis. The corpus does not store the first response before a retry.

The studied rule is action-conditioned. It blocks a demand increase only when the agent reports both high or very high water scarcity and high or very high coping capacity. Coping capacity is the agent's reported ability to adapt. This is the WAGF policy branch named `high_threat_high_cope_no_increase`; it is not a conclusion inferred from the report. High scarcity alone produces a warning, not a block. The block therefore makes the increase ineligible under this testbed policy for that reported pair; it does not claim that high scarcity alone warrants intervention.

We study this branch because the same agent-authored fields can enter both runtime enforcement and the measurement denominator. In the anchor irrigation cell, both conditions appear together in 25.46% of governed decisions and 68.61% of no-validator decisions, a $+43.15$ pp contrast. The primary conditioning variable is the scarcity report. It is broader than the two-condition rule: it

includes decisions satisfying both conditions and decisions with low coping capacity. We use this scarcity condition as the primary metric because it is the condition in the frozen analysis. A *world state* is recorded by the simulator rather than written by the agent, such as reservoir elevation or shortage tier.

Validator rules include both self-report and world-state checks. Figure 2a groups the 6,469 validator-on rejections in the anchor cell by the inputs read by the named rules. The "reads agent report" segment includes any rejection naming a rule that reads a reported condition; the "reads recorded world" segment includes only simulator-recorded inputs. These segments account for 59.4% and 40.6% of the rejections, respectively. Figure 2b instead uses all decisions as its denominator and compares the blocked share across models. Its on bars include both rule layers, whereas its off bars retain only the water-right cap, a separate world-feasibility check on increase proposals. The high-scarcity/high-coping increase rule belongs to the agent-report layer. This classification does not imply that every high-scarcity report is blocked.

2.2 Paired simulations and quantities we compare

Irrigation is the primary simulation setting. It contains 78 farm agents, 42 simulated years, and 3,276 decisions per run. Because the reservoir is coupled to the agent actions, an action in one year can affect later world state. Five model configurations are evaluated in irrigation. Their validator-on and validator-off runs are paired by seed; most cells use five seeds and two use three. Flood contains nine additional configurations and has no action-to-environment feedback. We use Flood only as a quantitative, descriptive cross-testbed check. In both settings, the high-threat-appraisal contrast uses the high/very-high labels from a shared five-level appraisal instrument; irrigation stores it as `WSA_LABEL` and Flood as `TP_LABEL`. The prompts, action ontology, feedback structure, and seed matching still differ. The corpus contains fourteen configurations with publicly available weights spanning 3B to 27B.

For decision i in seed s and arm a (validator-on or validator-off), let $R^a_{is} = 1$ indicate an agent report of high or very high scarcity. Let $A^a_{is} = 1$ indicate a demand increase in the EXECUTEDaction. With N_s decisions in a seed, we report:

$$P^a_{\text{high},s} = \frac{1}{N_s} \sum_i R^a_{is}, \tag{1}$$

$$r^a_{\text{cond},s} = \frac{\sum_i A^a_{is} R^a_{is}}{\sum_i R^a_{is}}, \tag{2}$$

$$r^a_{\text{all},s} = \frac{1}{N_s} \sum_i A^a_{is}. \tag{3}$$

A reported cell value is the mean of the per-seed values above. The contrast is always $\Delta = \text{off} - \text{on}$, so a positive value means that validators lowered the reported quantity. We report contrasts in percentage points (pp), using the seed as the unit for inference. The validator-on/off contrast is the object being audited, not an estimate of a governance effect: this is the contrast that governed-agent evaluations report, and governance changes the recorded world by design. The comparison is descriptive rather than causal by construction. The all-decision rate below supplies a denominator that governance cannot select. We use a two-sided one-sample t-test on the per-seed off-minus-on deltas; 95% confidence intervals use the corresponding t critical value with $n - 1$ degrees of freedom. Cellwise p-values are unadjusted for multiplicity, so the five-cell $p < 0.05$ summary is descriptive rather than a familywise claim. The LAST-PROPOSED reading is computed only as an explicit sensitivity check from the same released rows.

3 The Denominator Moves

3.1 Reported conditions and conditional rates

The anchor irrigation cell makes the denominator problem visible. High scarcity appears in 37.08% of decisions with validators and 82.80% without them. The share contrast is $+45.73$ pp (95% confidence interval (CI) $[+40.24, +51.21]$, paired $p < 0.001$). Among decisions reporting high scarcity, the

Table 1: Irrigation condition shares and executed-action rates. HIGH is the share of decisions with a high or very high agent-reported scarcity condition. The executed conditional increase rate is computed only within HIGH decisions; Figure 4 compares its contrast with the executed all-decision contrast. Values are means over paired seeds (n) using the action that executed after validation and any retry. Δ is no-validator minus validator-on.

Model	n	HIGH share (%)			Executed conditional increase (%)		
		on	off	Δ	on	off	Δ
gemma3:4b	5	37.08	82.80	+45.73	0.86	26.77	+25.90
gemma3:12b	3	5.49	26.76	+21.27	0.17	13.13	+12.96
gemma4:e2b	5	18.26	65.79	+47.53	0.00	27.96	+27.96
gemma4:e4b	5	45.27	88.65	+43.38	0.11	40.06	+39.95
ministral3:8b	3	13.56	44.69	+31.13	1.65	5.85	+4.20

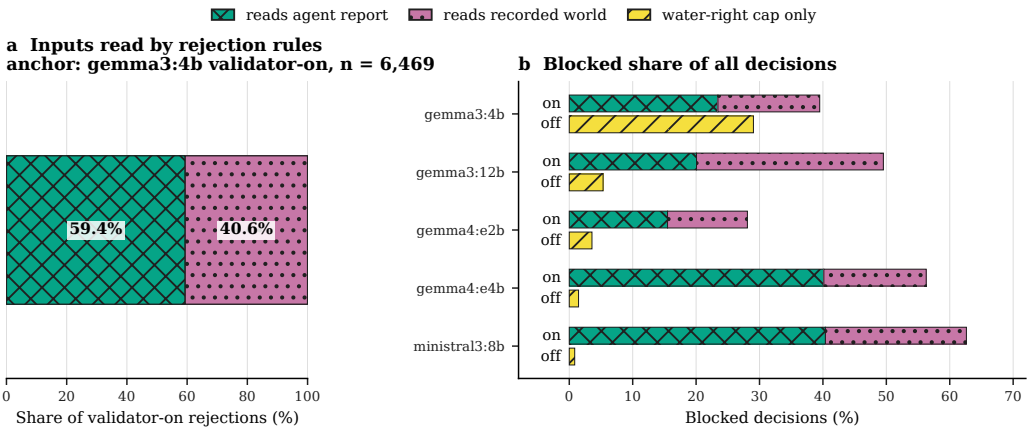


Figure 2: **Rule-layer decomposition.** Panel (a) groups 6,469 anchor-cell rejections by whether the named rules read an agent report or only the recorded world. Panel (b) uses all decisions as its denominator. Its on bars include both rule layers; its off bars retain only the water-right cap. High scarcity alone is a warning, not a block. The figure classifies rule inputs rather than estimating an effect.

executed increase rate is 0.86% with validators and 26.77% without them. This is a $+25.90$ pp contrast (95% CI $[+24.58, +27.23]$, $p < 0.001$). The executed rate and its denominator move together.

The five irrigation configurations all have positive share contrasts ($+21.27$ to $+47.53$ pp; all unadjusted cellwise p-values are below 0.05). Executed conditional contrasts range from $+4.20$ to $+39.95$ pp, versus $+20.26$ to $+36.57$ pp for all decisions. Table 1 places the condition share beside the conditional rate; a single score would hide that both the measured population and executed rate move.

3.2 Why this is a measurement problem

The conditional rate is not the fraction of all decisions that violate a rule. It is the rate of increase among decisions selected by an agent-authored condition. If validators alter the condition, the numerator and denominator no longer refer to a stable set of decisions. The reported conditional score alone cannot distinguish a change in the condition, a change in the executed action, or both.

4 World-State Composition Diagnosis

The coupled irrigation environment allows a descriptive composition check. An accepted increase changes the reservoir, and reservoir state enters later context. For each comparable recorded-world stratum, we compute each arm's high-scarcity share and give both arms the pooled stratum weights.

We require at least 40 records per arm; coverage is the share of an arm's decisions in retained strata. This asks how much of the raw contrast remains after aligning observed states. It is a descriptive composition diagnostic, not a causal adjustment, because the recorded state is downstream of governance.

Using the anchor cell, the raw share contrast is $+45.73$ pp. Reweighting by shortage tier leaves -2.53 pp, with 100% coverage in both arms. A finer grouping of recorded states leaves $+0.21$ pp, with 43.4% coverage in the governed arm and 36.4% in the no-validator arm. These are pooled estimates after reweighting both arms to the same recorded-world distribution. Table 2 and the supplement provide the per-seed diagnostics, which use strata with at least eight records per arm and report a paired interval only when at least three seeds retain comparable strata. The residual is small relative to the raw contrast, but this analysis does not show that an agent would report the same condition in an identical world. Finer groupings trade coverage for resolution.

The environment also diverges. Figure 3a labels year 1 as "matched worlds" because the arms match on nine pre-decision columns across 390 paired agent-years before actions separate their trajectories. The year-1 placebo check finds a $+3.33$ pp share contrast (95% CI $[-3.75, +10.42]$, $p = 0.261$); its interval includes zero, so the frozen data show no detectable baseline share difference. The yearly contrast is non-monotonic: it is negative in year 4 (-18.46 pp) and year 7 (-6.41 pp) before later positive values. Figure 3b marks the simulator's 895-ft floor. The no-validator arm reaches that floor in all five seeds, whereas the validator-on arm ends 106.57 ft higher on Lake Mead and has a 102.25-ft average gap over 42 years. Later separation makes recorded-world composition the relevant diagnostic.

The never-retried subset is a limited sensitivity check. It contains decisions for which no validator rejection message was returned. Its anchor contrast is $+53.46$ pp, larger than the all-decision contrast. The arms are filtered differently: 36.2% of governed decisions and 64.6% of no-validator decisions are never retried. Because governance affects retries, this filtering can change the compared population; the subset is not evidence that selection is irrelevant.

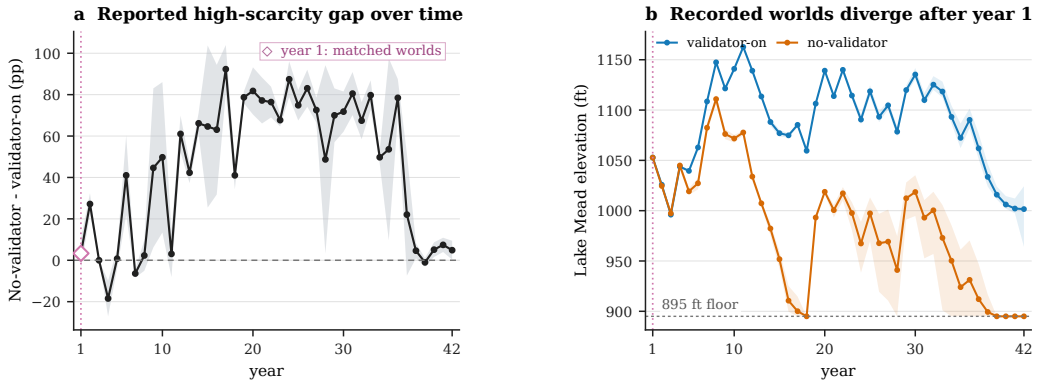


Figure 3: **Primary irrigation world-state diagnosis.** Both panels use the anchor cell and five paired seeds. Panel (a) reports the yearly no-validator-minus-validator-on difference in high-scarcity share; gray shading is its paired 95% confidence interval. The diamond marks year 1, when pre-decision worlds match. Panel (b) reports mean Lake Mead elevation; colored bands span the five seed values and the dashed line marks the 895-ft floor. Table 2 supplies the composition diagnostic; these panels do not identify causal mediation.

5 Different Denominators and Action Readings

Runtime governance keeps behavior within modeled rules, which makes the environment interpretable. The measurement risk begins when the same agent-authored input also selects the scored population. The conditional rate asks how often the executed action is an increase among high-scarcity decisions; the all-decision rate asks the same question across every decision. They answer different questions, and neither is an automatic correction. Because a rejected proposal can remain in a legacy field even when a default action executes, we use the executed action as the primary reading.

Table 2: Selected world-state reweighting diagnostics for the anchor cell. The pooled difference is the no-validator minus validator-on share after reweighting both arms to the pooled recorded-world distribution. Coverage is reported as governed / no-validator. The per-seed diagnostic is a separate estimate with a paired 95% interval.

Stratification	Coverage (on/off)	Pooled difference (pp)	Per-seed (95% CI)
Shortage tier	100.0/100.0%	-2.53	-3.12 [-5.56,-0.68]
Tier + Mead elevation + drought + agent cluster	43.4/36.4%	+0.21	-0.87 [-4.94,+3.19]

Using executed actions, the anchor conditional contrast is $+25.90$ pp and the all-decision contrast is $+29.90$ pp. Across all five irrigation cells, the executed all-decision contrast ranges from $+20.26$ pp to $+36.57$ pp (gemma3:12b: $+27.82$ pp; ministral3:8b: $+20.26$ pp). Figure 4 uses a blue circle for the conditional contrast among high-scarcity decisions and an orange square for the contrast across all decisions; gray lines connect these point estimates, and colored horizontal bars show paired 95% intervals. The right-hand value is conditional minus all decisions, with a paired 95% interval; a negative value means that conditioning produces the smaller contrast. This gap ranges from -16.06 pp to $+3.38$ pp: four cells are smaller conditionally and one is larger. Legacy LAST-PROPOSED values for gemma3:12b are $+11.44$ pp conditional and $+0.16$ pp all-decision; for ministral3:8b they are $+2.80$ pp and -22.17 pp. These are sensitivity values, not the primary estimand.

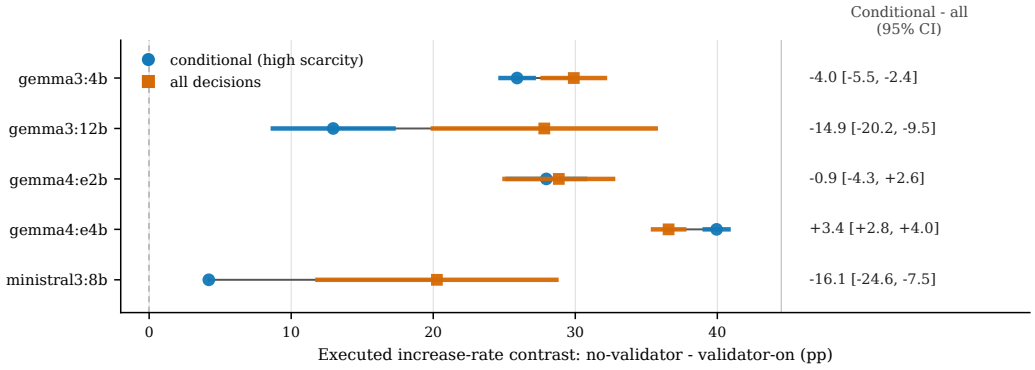


Figure 4: **Executed-action denominator gap.** Each row connects no-validator-minus-validator-on contrasts computed from executed increases. Blue circles use high-scarcity decisions; orange squares use all decisions. Gray lines connect point estimates; colored lines are paired 95% confidence intervals. The right column reports conditional minus all decisions, so a negative value means that conditioning produces the smaller contrast. Four gap intervals exclude zero; the gemma4:e2b interval includes zero. Neither endpoint is an automatic correction.

5.1 SECONDARY CROSS-TESTBED BOUNDARY CHECK

The nine Flood configurations span -3.66 pp to $+6.24$ pp, whereas irrigation spans $+21.27$ pp to $+47.53$ pp. Flood governance rejects 0–0.64% of proposed actions, compared with 28–63% in irrigation. Table 3 reports this separation. Flood differs in field names, action ontology, feedback, and seed matching; manifest seed IDs differ across arms in six of nine configurations, and minimum audit coverage is 70.5–99.9%. We use Flood only as a broad non-causal boundary check, not a clean control, dose-response estimate, or mechanism-transfer test. We do not use these configurations to support causal claims. Flood rows with a blank appraisal after retry exhaustion remain in the denominator; the settled B6 exclusion is documented in the table and the machine-readable provenance file.

Table 3: **Secondary cross-testbed boundary check.** Values are ranges across configurations; each configuration is the mean of per-run means. Δ is no-validator minus validator-on. Both settings use high/very-high labels from a shared five-level appraisal instrument. Flood is a bounded non-causal check, not a paired comparison, dose-response estimate, or mechanism-transfer test. Coverage is 70.5%–99.9%; manifest seed IDs differ across arms in six configurations. Blank appraisals on retry-exhausted Flood rows remain in the denominator. The Flood gemma3:4b cell excludes one governed run with a divergent prompt (four governed and five no-validator runs); including it changes that cell from $+0.35$ pp to -1.16 pp.

Setting	Configurations	High-threat share on	High-threat share off	Contrast range	Mean Δ
Irrigation	5	5.49–45.27%	26.76–88.65%	$+21.27$–$+47.53$ pp	$+37.80$ pp
Flood	9	2.55–43.48%	0.93–49.71%	-3.66–$+6.24$ pp	$+0.75$ pp

6 Reporting Protocol and Limitations

The reporting protocol has five items. Report (1) the joint share satisfying all agent-authored conditions in both arms, (2) the corresponding simulator state when one exists, and (3) an all-decision rate when the design defines one. Also store (4) the first response and validation message before a retry and (5) a state trajectory when actions change the environment.

Replacing the agent report with a recorded condition is not an automatic fix. The anchor shortage-tier contrast is $+64.29$ pp, versus $+45.73$ pp for reported scarcity and $+43.15$ pp for the two-condition rule. The simulator record enables reweighting but is changed by governance through the reservoir loop, so it defines a different quantity.

The main limitation is missing first-attempt information. About 24.3% of governed anchor decisions are accepted after a retry, but the preceding report is absent. We cannot separate within-decision revision from selection into the recorded response.

The world-state decomposition covers one irrigation cell, and its state is downstream of governance. The publicly weighted model set contains no frontier models, while the simulation and rule regime are confounded. Our claim therefore concerns this coupled governance loop rather than all LLM agents. Flood remains a descriptive check because its source history and pairing differ; Table 3 records the B6 exclusion and unequal run counts.

7 Conclusion

Runtime governance keeps an agent within the rules that give its environment meaning. In a coupled simulation, it can also change which decisions enter a conditional score.

In the irrigation simulations, the condition-share contrast is $+45.73$ pp; reweighting to the same recorded world leaves -2.53 pp. Executed all-decision contrasts remain positive across five configurations, while conditional-minus-all gaps range from -16.06 pp to $+3.38$ pp. Four paired gap intervals exclude zero. A simulator-recorded condition is not an automatic repair; its anchor contrast is $+64.29$ pp.

We treat the validator-on/off contrast as an object of study rather than a governance-effect estimate. Evaluations should compare matched world states when possible and report both a fixed-denominator rate and the condition share. Otherwise, a conditional score measures the agent together with the governance that selects its records.

AI Use Statement

AI assistants supported study planning, review, interpretation of released outputs, manuscript editing, evidence bookkeeping, and generation of the non-data-bearing Figure 1. Reported numbers came from documented scripts over the frozen corpus. The authors reviewed the AI-assisted code, prose, citations, claims, and figure labels and take responsibility for the submission.

Ethics Statement

The study uses simulated agents and does not recruit human participants. The submission excludes private credentials and identifying run metadata. Future source-profile or manifest releases will be limited to artifacts cleared for redistribution and anonymization.

Reproducibility Statement

The paper reports model configurations, paired arms, seed-level inference, action readings, and metric definitions. The supplement provides frozen values, regeneration commands, and missing-field records; historical evidence scripts remain outside the release package.

The inventory contains 42 historical runs. Legacy Git fields were recorded at finalization rather than launch, and 15 runs lack a recorded commit. Worker evidence is incomplete or mismatched for the five gemma3:4b pairs. Ten of 21 pairs meet stricter observable-alignment criteria; the remaining pairs retain complete analysis artifacts but limited provenance. The paper does not claim bitwise reproduction.

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